

in the track, and then clicking and dragging the weight curve. Time Warps allow the animation in a clip to be squashed and stretched over time, allowing a motion to be retimed.

8.6 Physique

Physique, similar to the Skin modifier, attaches mesh to a skeleton such as a Biped. When you animate the skeleton with skin attached, Physique deforms the skin to match the skeleton's movement. Physique works on many objects, including geometric primitives, editable meshes, patch-based objects, NURBS, and even FFD space warps. Physique has several advanced features, such as the ability to create bulges, skin sliding, the amount of twist, and crease blending as a character moves. Physique's envelopes are much more customizable than skin, but it does not have Skin's sophisticated vertex weighting or painting and cannot do morph-based deformations.

8.6.1 Applying Physique

Before Physique is applied, the Biped needs to be in Figure mode. As with most characters, it's best to use a pose with the arms outstretched so the hands are away from the torso. This makes initial envelope assignment



Physique makes the skin deform with bone movement

easier. Apply the Physique modifier to the mesh. The main level of the Physique modifier has a tool called Attach To Node, which attaches the modifier to the root node of the hierarchy. In the case of a Biped, the root node is the Pelvis, not the COM. The COM actually moves as the Biped maintains balance, so that can cause unwanted bulging if it were included in Physique. Once

attached, Physique walks the hierarchy and adds all the attached joints. This works differently than Skin because Skin accepts any object as a bone, regardless of hierarchy. To add bones or joints outside of the hierarchy, use Physique's Floating Bones rollout.

After Physique has been initialized, adjustment of vertex weights and envelopes can begin. The process is usually done by adjusting vertices and envelopes first, and then moving on to bulges and tendons. The best way to test how the skin reacts is by animating the joints of the character so you